Gowri Arun

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Education

Indian Institute of Information Technology, Kottayam *Expected Jun 2028*
B.Tech. in Computer Science and Engineering CGPA: **9.11/10.00**

Technical Skills

Languages: Python, C++, TypeScript, JavaScript, SQL

Frameworks: FastAPI, Flask, Node.js, Express.js, React, PyTorch

Databases & Tools: PostgreSQL, SQLite, DuckDB, Supabase, Firebase, FAISS, Neo4j, Docker, Git, GitHub Actions

Experience

Research Intern | NIT Tiruchirappalli *May–Jul 2026*

- Reconstructed a dual-stream **PyTorch STHF model** for visible–infrared video re-identification, integrating 3D FFT filtering, ResNet-50, and spatial–temporal feature refinement.
- Designed an **adaptive three-branch frequency filter** with learnable sequence-specific gating; implemented modality-aware sampling and cross-camera retrieval evaluation.

Student Researcher | IIIT Kottayam *Jan 2026–Present*

- Built a leakage-controlled intrusion detection pipeline for **2.54M UNSW-NB15 records** using mutual-information feature selection, PCA, class balancing, and stratified cross-validation.
- Evaluated ML/deep-learning models; shared-feature BiLSTM achieved **98.70% binary accuracy / 95.10% F1**, with **96.66% multiclass accuracy** from a DNN.

Projects

Astra-Q [**Code**] [**Live**] | *FastAPI, FAISS, Neo4j, Gemini* *Dec 2025*

- Built MOSDAC satellite-document retrieval infrastructure with a **6,347-vector FAISS index**, intent routing, knowledge-graph queries, and source-linked responses.
- Eliminated duplicate retrieval and aligned displayed sources with generation context, reducing mean local mocked-generation orchestration latency **44.4% (90.3 → 50.3 ms)** over 150 runs.

NeuroBridge [**Code**] [**Live**] | *React, FastAPI, Supabase, PostgreSQL* *2026*

- Engineered deterministic intervention workflows and confidence-gated personalization memory, preventing **1,000 premature writes** across 200 synthetic streams versus an ungated baseline.
- Resolved a lifecycle self-deadlock and enforced append-only event integrity; cut redundant repository writes **25%** in a synthetic replay benchmark.

LazyLift [**Code**] [**Live**] | *React, TypeScript, Express, SQLite, Docker* *2026*

- Co-developed a study platform that extracts source-linked questions and priorities from course materials and builds constraint-aware schedules.
- Optimized PPTX slide rendering with SHA-256 caching, cutting median endpoint latency **99.83% (3.49 s → 5.94 ms)**; added in-flight deduplication for concurrent requests.

Veridian AI [**Code**] | *Python, DuckDB, SQL, BM25* *2026*

- Built a deterministic natural-language analytics pipeline with typed semantic parsing, parameterized SQL, read-only execution, and inspectable query traces.
- Raised exact-result correctness from **76.3% to 92.1% (+15.8 pp)** on 38 reference-checked queries by correcting intent classification and semantic parsing.

Achievements

- **Top 10 of 7,100+ registrations**, NMIT Hacks 2026; **First Runner-Up**, Girlathon 2025 (Kerala Startup Mission).
- **Top 10 Team**, She Builds Tech 2.0; Global Rank **3,592 / 13,000+**, IICPC Codefest Prelims 2026.